Online Appendix

Semiparametric Value-at-Risk and Expected Shortfall with a Real-Time Misspecification Score

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OA.1 Glossary of acronyms

ARCD	autoregressive conditional density model (Hansen, 1994)
CAViaR	conditional autoregressive value-at-risk (Engle and Manganelli, 2004); SAV, its symmetric abso
CC, UC	conditional coverage (Christoffersen, 1998) and unconditional coverage (Kupiec, 1995) exceptio
CRPS	continuous ranked probability score
CRSP, WRDS	Center for Research in Security Prices; Wharton Research Data Services
DM	Diebold–Mariano test (Diebold and Mariano, 1995)
EM, IG, OAS	emerging markets; investment grade; option-adjusted spread
ERM	empirical risk minimization
ES, VaR	Expected Shortfall; Value-at-Risk
EVT, GPD	extreme-value theory; generalized Pareto distribution
EWMA	exponentially weighted moving average variance (J.P. Morgan/Reuters, 1996)
FHS, HS	filtered historical simulation (Barone-Adesi et al., 1999); historical simulation
FRTB	Fundamental Review of the Trading Book (Basel Committee on Banking Supervision, 2019)
FWER	family-wise error rate
FZ0	the zero-homogeneous Fissler–Ziegel joint (VaR, ES) scoring function (Fissler and Ziegel, 2016;
GARCH, GJR	generalized autoregressive conditional heteroskedasticity (Bollerslev, 1986); the Glosten–Jagann
GAS	generalized autoregressive score model (Patton et al., 2019)
GBC	generative Bayesian computation (Polson and Sokolov, 2023)
GBM	gradient boosting machine (gradient-boosted regression trees)
GFC	the 2008 global financial crisis
IQN	implicit quantile network (Dabney et al., 2018)
MCS	Model Confidence Set (Hansen et al., 2011)
PIT	probability integral transform
PZC	Patton, Ziegel and Chen (Patton et al., 2019)
QML	quasi-maximum likelihood
RV	realized variance
SPA	superior predictive ability test (Hansen, 2005)
TAQ	NYSE Trade and Quote database

OA.2 Algorithms

Algorithm OA.1 (amortized shape model). (i) Assemble the pooled panel $\{(s_{j,t}, \widehat{z}_{j,t})\}$ over names j and days t. (ii) Train H_ϕ by SGD on $\mathbb{E}_{\tau\sim\lambda}\,\rho_\tau(\widehat{z} - H_\phi(s,\tau))$ [network], or boost per-τ regressions on a grid [trees]; enforce monotonicity by rearrangement. (iii) For any asset, including one with no return history, using characteristics-only s, evaluate $H_\phi(s_t,\cdot)$; no per-asset estimation step exists. (iv) To *sample*, draw $\tau^{(m)} \sim U(0,1)$ and return $z^{(m)} = H_\phi(s_t, \tau^{(m)})$.

Algorithm OA.2 (the misspecification meter, as deployed). (i) Fit equation (2) of the paper on a trailing window; store $\widehat{z}_{t-62}, \ldots, \widehat{z}_t$. (ii) Compute mk_{63}, $|\mathrm{sk}_{63}|$, J_5 by equation (9) of the paper. (iii) Convert each to its percentile against the trailing pooled panel (same asset class); the score is the maximum of the three percentiles, re-ranked into deciles. (iv) Act by the paper: top decile $\Rightarrow$ the parametric model's average pinball loss runs ∼2–3% higher within this score region (a conditional average, not a per-name point estimate), flag the asset for the nonparametric / residual-hybrid engine and for model-risk review; elsewhere $\Rightarrow$ no action (in the amortized setting the engine is run regardless; the meter serves surveillance, and day-level switching is unsupported, Section 4.3 of the paper). (v) Latency: detection lags episode onset by median 0 / mean 2–4 days; the edge is back-loaded within episodes, so the lag forfeits little.

Deployed pipeline. Each day: (i) update each name's GARCH filter state $\widehat{\sigma}_{t+1}$ (recursion only; parameters refit annually); (ii) assemble the state vector s_t and evaluate the shared shape model $q_\phi(\tau \mid s_t)$, one model for the whole book, no per-name estimation; (iii) apply the frozen GPD tail below p_0 and, where the desk elects the coverage overlay, the conformal shift c_τ (static or adaptive); (iv) report $\widehat{Q}_{t+1}(\tau) = \widehat{\mu} + \widehat{\sigma}_{t+1}\left(Q^z_{t+1}(\tau) + c_\tau\right)$ at the desk's levels, with VaR and $\mathrm{ES}_{97.5}$ both read from the coherent min-envelope curve Q^z of equation (5) of the paper (VaR at the α node, ES its numerical tail integral); (v) read the misspecification score equation (9) of the paper as the model-risk monitor: top decile means the parametric fallback's conditional average loss in that score region runs ∼2–3% higher. *Annually:* refit filters, retrain the shape model, re-estimate tail and shifts, all on trailing data. *New listing:* run the separate characteristics-only amortized forecaster until an own-history sample sufficient to initialize the per-name scale filter accrues; then attach this residual-hybrid pipeline. Total marginal cost per name per day: one filter update and one model evaluation.

OA.3 Supplementary figures

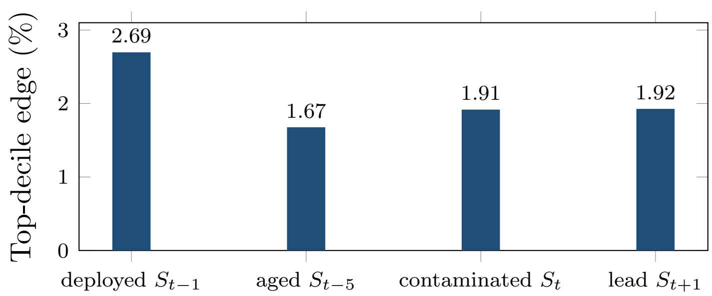

Figure OA.1: Timing diagnostics for the frontier. Top-decile edge under four alignments of the mk_{63} score with the daily loss (200 CRSP names; per-date DM 5.5–6.5 in all four). The deployed lag dominates, and the contaminated same-day alignment yields less, the opposite of what a look-ahead artifact would produce.

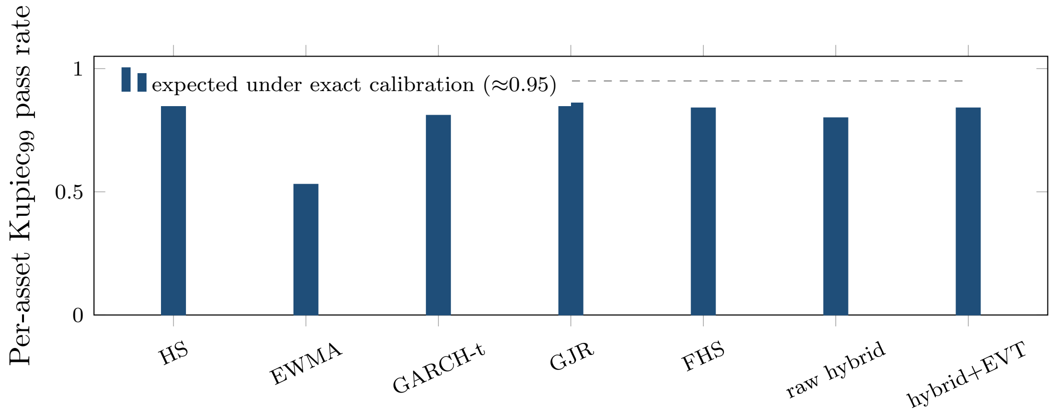

Figure OA.2: Exception tests restated per asset (140 CRSP names, 99% VaR): share of names whose individual Kupiec test is consistent with nominal. HS over-covers (conservative); EWMA fails; the EVT-tailed engine matches the parametric benchmarks while beating them on loss and the joint FZ0 score (Table 6 of the paper). Date-clustered tests in the table notes.

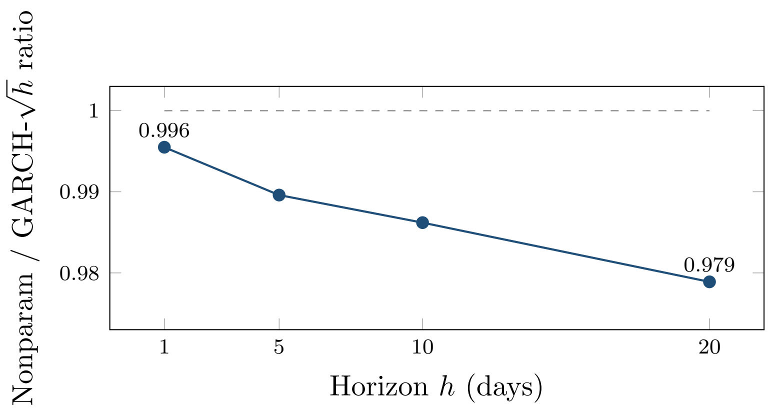

Figure OA.3: The edge grows with horizon. Direct-model/GARCH pinball ratio at $h = 1, 5, 10, 20$ days (150-name design panel; training labels whose h-day window crosses the split are purged): fat tails compound while $\sqrt{h}$ scaling compounds the shape error, from 0.45% at h=1 to 2.1% at h=20. The forecaster is the direct multi-day model of the main text's ten-day subsection, *not* the residual-hybrid engine, whose architecture is one-day.

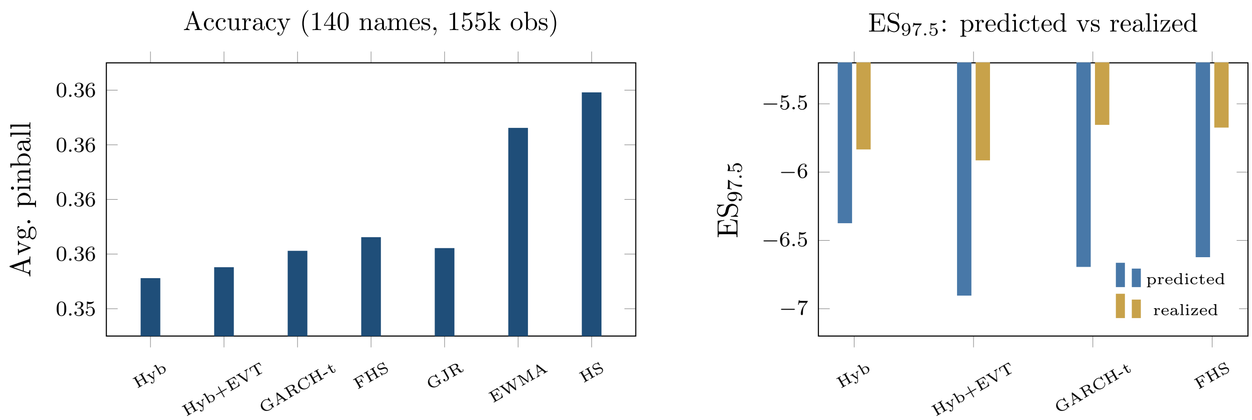

Figure OA.4: The FRTB battery. Left: average pinball loss by model (axis truncated to resolve the ordering; differences carry DM 5.2–8.8 vs the hybrid, all $p \approx 0$); CAViaR, not shown, statistically ties the hybrid. Right: $\mathrm{ES}_{97.5}$ predicted (exact tail integral) vs realized below each model's own VaR, a model-dependent diagnostic, not a ranking (every fat-tailed entrant's integral exceeds its own-tail mean); the joint FZ0 score of the paper is the ranking criterion.

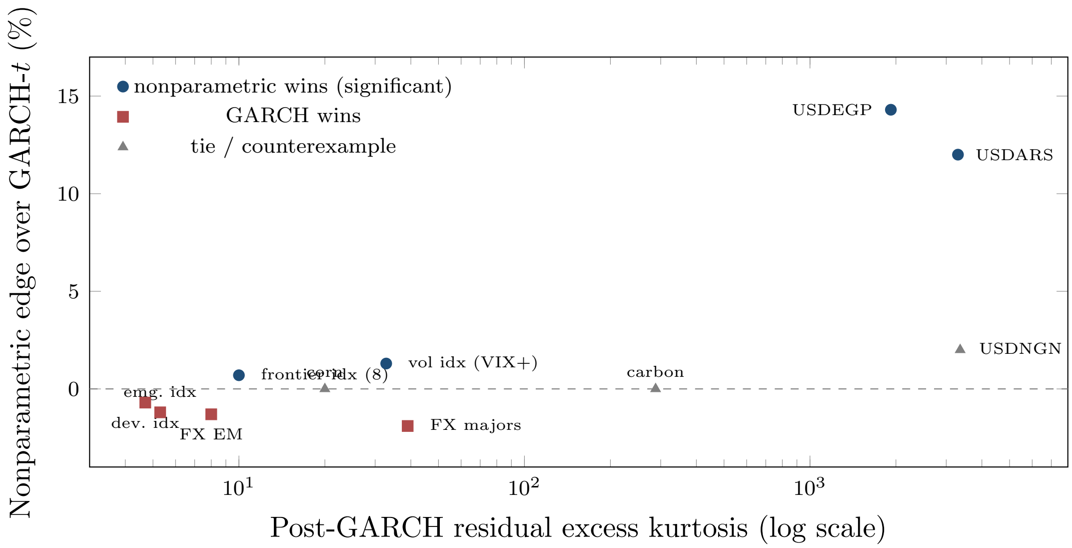

Figure OA.5: The frontier across universes: nonparametric edge vs post-GARCH residual kurtosis, tier means and single assets with study-reported kurtosis only (country universe corr 0.53; cross-asset corr 0.43). The hyperinflation-FX tier (kurtosis ∼2000–3300) carries the only decisive standalone wins; high kurtosis is predictive but neither necessary nor sufficient: carbon (288) and corn (20) tie, and NGN's win is not significant. The Korea-2026 control (kurtosis 4–13, GARCH wins all 23 assets) sits in the lower-left regime.

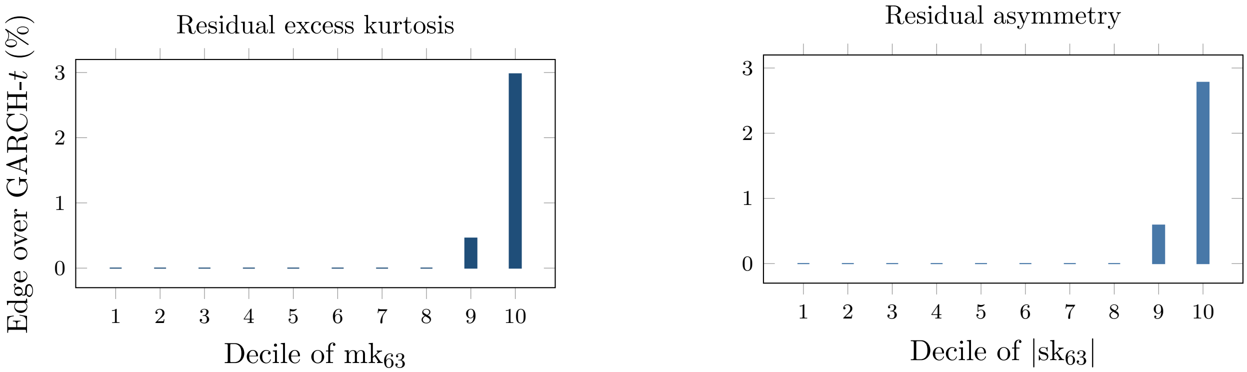

Figure OA.6: The misspecification frontier. Nonparametric pinball edge over GARCH-t by decile of the trailing 63-day residual excess kurtosis (left) and residual asymmetry (right), 200 CRSP names, 221.6k test asset-days. Deciles 1–8 are ≈ 0 (drawn at zero; exact bulk values in the result file); the edge jumps in the top decile (+2.98% kurtosis, +2.78% asymmetry; per-date DM 10.5 in the top kurtosis decile).

OA.4 Supplementary tables

Direct expected-shortfall backtests

Exception counts test the VaR level; they do not test ES. The main text adds two direct ES *calibration* backtests: the McNeil–Frey (2000) exceedance-residual test, whose statistic is the mean standardized loss beyond VaR, zero under a correct ES (McNeil and Frey, 2000); and the Acerbi–Székely (2014) statistic Z_2, negative when ES understates the realized tail (Acerbi and Székely, 2014). These complement the jointly consistent FZ0 loss used for forecast comparison; Nolde and Ziegel (2017) and Bayer and Dimitriadis (2022) draw the distinction between backtesting the calibration of an ES forecast and comparing forecasts through a jointly consistent score. Table OA.5 reports the two calibration tests on the same 200-name panel and 221,600 held-out asset-days (1,108 test dates) as the full-panel FZ0 re-scoring, with the engine's VaR and ES read from its own coherent min-envelope curve Q^z of equation (5) of the paper — the exact forecasts the joint loss scores, not a simpler proxy — and inference by a stationary block bootstrap over calendar dates. Gaussian innovations are decisively inadequate in this daily-data battery: the Gaussian filter is worst on every ES metric, its mean exceedance residual reaching -0.77 standard deviations at the 1% level ($Z_2 = -0.99$). Among the fat-tailed entrants no model dominates every diagnostic. On the Acerbi–Székely Z_2 the engine is best calibrated in the tail: it carries the statistic closest to zero at 1% (-0.07, not rejected) and is effectively tied with GARCH-t at 2.5% (-0.11 against -0.10). The engine and GARCH-t are the two entrants that pass the 1% Kupiec unconditional-coverage test; FHS and the Gaussian filter fail it. On the McNeil–Frey exceedance residual FHS — which fits the empirical residual tail directly — sits closest to zero and is the only entrant the test does not reject at 1% (-0.10, $p = 0.07$); at 2.5% FHS, GARCH-t and the engine all fall inside the non-rejection region. The reading is consistent with the exception tests and the joint score: fat tails materially improve tail calibration, the engine and GARCH-t are the best-calibrated pair on the direct tests, and the engine's decisive margin is the comparative FZ0 loss, where it beats both GARCH-t and FHS at each level.

Realized measures and the scale–shape decomposition

The main-text battery reads only daily returns, the regime the estimator is built for. To place it against a method that reads intraday data, and to ask what the misspecification frontier is made of, we build five-minute realized measures from TAQ for thirty large-capitalization U.S. equities over 2014–2024 (realized variance is the sum of squared five-minute log-returns, cleaned by regular sale conditions, a daily price band, and a 20% return clip) and fit a Realized GARCH (Hansen et al., 2012): the log-linear conditional variance $\log h_t = \omega_R + \beta_R \log h_{t-1} + \gamma_R \log \mathrm{RV}_{t-1}$ with a measurement equation $\log \mathrm{RV}_t = \xi_R + \phi_R \log h_t + \tau_R(z_t) + u_{R,t}$ linking the realized measure to the

Table OA.1: Summary of the paper's results and scope conditions. Edges are percentage pinball reductions unless noted; DM statistics are per-date (conventions note of Section 3 of the paper), one-sided. Panel A: the deployed amortized residual-hybrid engine (worst case on the accuracy yardstick: no detectable loss; calm-cell 90% lower confidence bounds sit above −0.25%). A separate block reports the direct multi-day extension, which is not the engine. Panel B: boundary evidence from configurations not deployed, which motivates the design.

Panel A: the engine Result	Magnitude	Significance
vs GARCH-t / EWMA / HS, h=1 (avg)	+0.3 / +1.5 / +1.9%	DM 4.4–8.3
Top misspecification decile	+2.98%	DM 10.5
Hyperinflation FX (ARS, EGP)	+12–14%	DM 2.9–3.8
Frontier equity indices (4 of 8)	∼+1–3%	DM 2.1–3.5
2000–2013 frozen-spec holdout (untouched era)	top decile +0.9%	DM 2.2 (sort); recursive real-time rule +1.0%, DM 2.15; frozen-constant restatement in text
Point-in-time universe (Q1-2000 top 200, delistings kept)	top bucket +2.9%; overall +0.6%	DM 6.6 / 9.0, frozen real-time thresholds
Strict calendar splits; annual-refit walk-forward	top-decile edge holds under all three	DM 6.2 (2020 cut); 3.1 (overall, pre-GFC cut); 4.41 (annual-refit residual-hybrid walk-forward, +2.12%)
Familywise error over the 3 × 10 signal-decile grid	9 cells survive 5% FWER	Romano–Wolf step-down, incl. all three top deciles
Exception battery through the 2008 crisis	Kupiec 94% / Christof. 95% at 99%	date-clustered t 0.17 at 99%; at 97.5%, 0.35 (EVT tail) and −1.09 (static overlay, conservative); HS, EWMA fail
Joint (VaR, ES) FZ0 score, full panel	accuracy layer lowest at both levels	DM 4.8–4.9 vs GARCH-t, FHS at 1%, 5.1 at 2.5%; 8.6–9.5 vs PZC GAS, 3.6-to-tie vs Taylor ES-CAViaR (multi-start); static shift concedes 2.5% (DM −2.0), adaptive shift erases it (tie; top-decile cost remains)
Cold start / young listings	+6–10%	full-scale panel; no per-asset model applies
Calm-quintile edge	≈ 0 (point estimates +0.0–+0.2%)	bottom-decile 90% CIs lie above −0.25%; no detectable loss
CAViaR (strongest academic rival)	tie (both in 90% MCS)	DM 0.4
Extension (a separate direct multi-day model, not the engine)		
Ten-day horizon vs $\sqrt{h}$-scaling	+1.4% (2014–24)	holds in 2020–22 stress; 2000–13 rerun: tie, DM 0.07 (era-dependent)
Panel B: boundary evidence from configurations not deployed		
Bare standalone neural net, single names	−4.5 to −6.2%	DM +12 to +18; motivates the hybrid design
Standalone nonparametric, calm single assets	GARCH as good or better	pooled DM −9.5 to −13.7; the shape edge needs misspecification
Korea 2026 crisis, 23 assets (falsification case)	GARCH wins all	DM −2.5 to −5; a price crash is not a shape break
mk_{63}-reweighted FHS (reweighting probe)	worse than plain FHS	DM −11 overall and top decile; sharp univariate conditioning collapses the sample
Baltic freight; IG credit[a]	+15%; +52%	DM 9.6; 30.6; accuracy only, calibration open

[a] Credit and freight fail breach-independence (exceptions cluster); their entries are accuracy results only. The credit magnitude is from the FRTB-grade single-asset battery; the smaller 4% figure in the paper's cross-universe table is from the separate lean hybrid-vs-GARCH sweep, different battery, both reported.

Table OA.2: Double sort on the misspecification score and trailing volatility. Cells are the nonparametric-over-GARCH-t pinball edge (%) in independent quintiles of the misspecification score (rows) and trailing volatility (columns), design-era test panel.

	rv_{63} Q1 (calm)	Q2	Q3	Q4	Q5 (active)
mk_{63} Q1 (low)	+0.24	+0.34	+0.10	+0.01	−0.14
Q2	+0.16	+0.20	+0.22	+0.09	−0.57
Q3	−0.02	+0.19	+0.25	+0.44	+0.04
Q4	+0.08	+0.15	+0.35	+0.23	+0.10
Q5 (high)	−0.10	+0.44	+0.68	+0.61	**+2.33**

Top-row date-clustered DM by volatility quintile: 0.7, 2.3, 1.2, 2.8, 6.4. The bottom-left and top-right regions are the point: neither ingredient alone produces the concentration.

Table OA.3: Model selection by task and regime. Each row is established in the section cited.

Task / regime	Winner	Where
Point quantile accuracy, tabular state	GBM (trees)	Section 2.3 of the paper
Single asset, calm (∼90% of asset-days)	GARCH-t	Section 4 of the paper
Single asset, top misspecification decile	Nonparametric (+2.98%, DM 10.5)	Section 4 of the paper
Amortized cross-name panel, any regime	Nonparametric always; no gate	Section 4.3 of the paper
Regulatory ES with a ≥GARCH floor	Residual-hybrid + EVT + conformal	Section 5 of the paper

latent variance, estimated jointly by Gaussian quasi-maximum likelihood. Table OA.6 reports the result on 1,108 common test dates. Two facts follow. The realized scale lowers FZ0 well below the daily-data accuracy core at both levels (DM 6.2 at 2.5%, 4.5 at 1%): intraday information prices volatility more sharply than any daily filter, and the economic gap is largest in the top misspecification decile, where the daily scale is coarsest. But the gain is scale, not shape. Holding the realized scale fixed and re-fitting the pooled state-conditioned residual-quantile learner, the EVT tail, and the misspecification score on the realized residuals moves FZ0 within noise: every stage's Diebold–Mariano statistic against the realized benchmark is below 1.4 in magnitude, overall and in the top score decile, the largest being the shape learner's −1.3 at 1% (directional, not significant). On these large-capitalization names a realized scale absorbs most of what the score detects on daily-GARCH residuals, so on the intraday-covered segment the two are substitutes; the score's distinctive value is in the daily-data regime and the cross-asset and day-one settings where a realized scale does not exist.

Table OA.4: Cross-universe multiplicity: Holm step-down over all 93 instrument-level comparisons in the three exploratory universes of the paper's cross-universe table (24 FX pairs, 26 country indices, 43 cross-asset instruments). One-sided p from each instrument's date-clustered DM statistic; familywise $\alpha = 5\%$. Ranks below twelve all have $p > 0.02$ and are omitted.

Rank	Instrument	Universe	DM	p (one-sided)	Holm cutoff	Verdict
1	PJM power[a]	cross-asset	283.1	$< 10^{-16}$	5.38×10^{-4}	artifact[a]
2	Baltic Dry	cross-asset	9.36	$< 10^{-16}$	5.43×10^{-4}	survives
3	VIX	cross-asset	5.71	5.7×10^{-9}	5.49×10^{-4}	survives
4	V2X	cross-asset	5.03	2.5×10^{-7}	5.56×10^{-4}	survives
5	VVIX	cross-asset	4.66	1.6×10^{-6}	5.62×10^{-4}	survives
6	USDEGP	FX	3.80	7.2×10^{-5}	5.68×10^{-4}	survives
7	Kenya NSE20	indices	3.52	2.2×10^{-4}	5.75×10^{-4}	survives
8	Wheat	cross-asset	3.51	2.2×10^{-4}	5.81×10^{-4}	survives
9	MOVE	cross-asset	3.22	6.4×10^{-4}	5.88×10^{-4}	fails
10	Sri Lanka CSEALL	indices	3.20	6.9×10^{-4}	5.95×10^{-4}	fails
11	Gasoline	cross-asset	3.20	6.9×10^{-4}	6.02×10^{-4}	fails
12	USDARS	FX	2.86	2.1×10^{-3}	6.10×10^{-4}	fails

The family is every instrument-level look in the three exploratory universes; the Korea-2026 falsification case (23 assets, GARCH-t wins throughout) is not part of the win family. Bonferroni at 0.05/93 gives the identical survivor set. The pooled crisis-FX comparison (DM 4.12, $p \approx 1.9 \times 10^{-5}$) and the kurtosis–edge correlations (0.53 across indices, 0.43 cross-asset) are pooled and gradient statistics outside this instrument-level family; they are the replication evidence the paper's cross-universe claims rest on.

[a] Retained in the family because the raw sweep contains it, but not a win: the paper's cross-asset audit reclassifies electricity as a log-return artifact on near-zero and negative prices, refit on price differences, GARCH wins PJM (DM −2.73), so the survivor count the text quotes is seven.

Table OA.5: Direct expected-shortfall backtests. 200 CRSP names, pooled test panel — the same held-out rows as the full-panel FZ0 re-scoring; the engine's VaR and ES are its own coherent min-envelope forecasts. Breach is the realized exception rate (target α); Kupiec p is the pooled unconditional-coverage test. Z_2 is the Acerbi–Székely (2014) statistic, negative when ES understates the tail. MF is the McNeil–Frey (2000) mean exceedance residual (standardized loss beyond VaR, zero under correct ES) with its bootstrap p.

	$\alpha = 2.5\%$				$\alpha = 1\%$			
Model	Breach	Kupiec p	Z_2	MF (p)	Breach	Kupiec p	Z_2	MF (p)
GARCH-normal	0.027	0.00	−0.31	−0.55 (0.00)	0.016	0.00	−0.99	−0.77 (0.00)
GARCH-t	0.027	0.00	**−0.10**	−0.05 (0.14)	0.010	**0.17**	−0.09	−0.23 (0.00)
GARCH-FHS	0.031	0.00	−0.24	**+0.01** (**0.84**)	0.012	0.00	−0.29	**−0.10** (**0.07**)
Engine (EVT-tailed)	0.027	0.00	−0.11	−0.04 (0.26)	0.010	**0.15**	**−0.07**	−0.14 (0.04)

Inference for the McNeil–Frey and Z_2 p-values is a stationary block bootstrap over the 1,108 trading dates (mean block ten dates) rather than treating the 221,600 asset-days as independent, since the 200 names share within-day market shocks. Kupiec is the pooled analytic unconditional-coverage test, hypersensitive at this sample size (every model is rejected at 2.5%). The deployed engine adds a 97.5% conformal shift and is conservative out of era (breach 1.66%, $Z_2 = +0.29$ at 2.5%), overstating the tail in the regulator-preferred direction. Source: `engine_esbt_results.json`.

Table OA.6: The scale–shape decomposition. Thirty large-cap U.S. equities, 2014–2024 (1,108 test dates), joint (VaR, ES) FZ0 loss (lower is better) at both FRTB levels. The scale is a proper Realized GARCH except in the first row (the daily-data accuracy core). DM is date-clustered (Newey–West, 5 lags) against the realized scale with an unconditional residual quantile; a positive DM is worse than that benchmark.

Model	FZ0 2.5%	DM	FZ0 1%	DM
Daily core (GARCH-t + EVT)	1.500	+6.2	1.774	+4.5
Realized scale + residual quantile	1.445	—	1.716	—
+ EVT tail	1.444	−0.3	1.716	+0.5
+ pooled shape learner	1.445	+0.2	1.710	−1.3
+ shape + EVT (realized-hybrid)	1.447	+0.7	1.718	+0.6

All five rows are computed in one pipeline on the same test rows, with ES the 20-node integral of each row's own quantile curve. The daily core's DM is against the realized benchmark (the intraday-scale gain). The pooled shape learner is the paper's state-conditioned residual-quantile model, retrained on the realized-scale residuals. Returns are CRSP daily; the intersection is common across models. Source: `scaleshape_canonical_results.json`.

OA.5 Standalone GARCH-EVT on the canonical rows

The manuscript's comparison set omits standalone GARCH-EVT (McNeil and Frey, 2000), which Kuester et al. (2006) rank first among classical methods. This section reports it on exactly the rows, filter and splits of the frontier sorts (Table 1 of the paper) and of the full-panel FZ0 run (Figure 2 of the paper), from `job_garch_evt.py` (result file `garch_evt_results.json`). Two forms are run: per name, with the GPD tails fitted to each name's own training residuals, and pooled, with one threshold and one (ξ, β) per tail across names. The pooled body row is the frontier engine of Table 1 (no EVT branch); the engine row adds the body/EVT minimum and rearrangement of equation (5). Table OA.7 gives the eleven-level pinball comparison and Table OA.8 the joint score.

Table OA.7: GARCH-EVT on the frontier rows: pinball edge by score region and head-to-head comparisons.

Model	Top mk_{63} decile	Deciles 1–9	Top composite decile	Overall
Edge over GARCH-t, % (per-date DM)				
GARCH-EVT, per name	+0.31 (3.62)	−0.22 (-6.61)	+0.37 (5.35)	−0.16 (-5.00)
GARCH-EVT, pooled tail	−0.16 (-2.69)	−0.12 (-4.55)	−0.06 (-1.40)	−0.12 (-4.89)
GARCH + pooled body (no EVT)	+2.98 (10.48)	+0.02 (0.43)	+2.79 (8.91)	+0.35 (5.35)
Engine (body/EVT minimum)	+2.48 (10.08)	−0.01 (-0.25)	+2.19 (8.00)	+0.27 (4.62)
Head-to-head, % (per-date DM)				
Engine over GARCH-EVT pooled	+2.64 (8.91)	+0.10 (2.47)	+2.24 (7.51)	+0.39 (7.66)
Engine over GARCH-EVT per name	+2.18 (9.63)	+0.21 (4.63)	+1.82 (7.39)	+0.43 (8.22)
Pooled body over GARCH-EVT pooled	+3.14 (9.39)	+0.14 (2.84)	+2.84 (8.37)	+0.47 (7.96)
Engine over pooled body	−0.51 (-9.04)	−0.04 (-2.28)	−0.62 (-9.64)	−0.09 (-5.25)

200 names, 221,600 test rows, same rows and splits as Table 1 of the paper and Figure 2 of the paper. Edge = (reference pinball − model pinball)/reference pinball over the eleven-level grid; DM is the per-date Newey–West(10) Diebold–Mariano statistic. GARCH-EVT is McNeil–Frey: GARCH(1,1)-t filter, GPD tails fitted by maximum likelihood to the 10% most extreme training residuals per tail (per name, or pooled across names), empirical residual quantiles between the thresholds. The engine's pooled GPD at $p_0 = 0.025$ has $\hat{\xi} = 0.298$, $\hat{\beta} = 0.639$ on 6075 exceedances; the EVT branch is the tighter of the two at the 1% node on 60.1% of test rows and at the 2.5% node on 59.5%.

Table OA.8: GARCH-EVT on the full-panel FZ0 rows.

Model	$\alpha = 1\%$ FZ0	$\alpha = 1\%$ Breach	$\alpha = 1\%$ DM vs engine	$\alpha = 2.5\%$ FZ0	$\alpha = 2.5\%$ Breach	$\alpha = 2.5\%$ DM vs engine
Engine (body/EVT minimum)	2.1473	1.04%	—	1.8494	2.78%	—
Engine + conformal overlay (97.5%)	2.1473	1.04%	0.00	1.8656	1.66%	2.29
GARCH + pooled body (no EVT)	2.1411	1.32%	-1.33	1.8433	3.25%	-1.87
GARCH-EVT, pooled tail	2.1551	1.06%	4.83	1.8583	2.89%	6.85
GARCH-EVT, per name	2.1693	1.13%	5.23	1.8596	2.93%	5.39
GARCH(1,1)-t	2.1557	1.03%	5.00	1.8542	2.71%	4.55

FZ0 is the zero-homogeneous Fissler–Ziegel joint (VaR, ES) loss, lower is better; DM > 0 means the row model scores worse than the engine (per-date Newey–West(10)). The engine row is the accuracy layer (no conformal shift) at both levels and is the reference of every DM; the overlay row adds the split-conformal shift at 97.5% and coincides with the engine at 1%. GARCH-EVT ES is the McNeil–Frey closed form; the engine and body ES are the 20-node integral of the monotonized quantile curve. Top-mk_{63}-decile DM of pooled GARCH-EVT versus the engine: 2.63 at 1% and 4.26 at 2.5%.

Threshold diagnostics. Over a threshold grid $p_0 \in \{0.025, 0.05, 0.10, 0.15\}$ the pooled lower-tail fit gives $\hat{\xi} = 0.298, 0.269, 0.204, 0.164$ on 6,075, 12,150, 24,300, 36,450 exceedances; per name at $p_0 = 0.10$ the interquartile range of $\hat{\xi}$ is 0.058 to 0.281 (median 0.174) with a median of 125 exceedances, and no name has $\hat{\xi} \geq 1$. The EVT branch is the tighter of the two inside the engine's minimum on 60.1% of test rows at the 1% node and 59.5% at the 2.5% node. On the 2000–2013 holdout rows (`job_holdout_garch_evt.py`) the frozen learner beats the pooled GARCH-EVT by +1.16% (DM 2.52) in the top holdout decile and by +1.30% (DM 1.38) in the frozen design-era bucket, and by −0.01% (DM -0.31) in deciles one through nine.

OA.6 Frontier robustness: calendar overlap, family-wise error, and the universe rule

This section collects the constructions summarized in Section 3 of the paper.

Calendar overlap and the pooled learner. The design panels are split per name, 60/40 in each name's own time index. In an unbalanced panel such a rule can place one name's training window on another name's test dates, so a pooled learner could meet a systemic event in training that a per-name model would meet only out of sample, the panel form of overlap leakage that purging and embargo address. Near-complete histories put the split points on almost the same date, so the channel is nearly degenerate here, and a strict calendar-time split closes it: each stage (each GARCH fit, the pooled learner, the GPD tail) is re-estimated strictly before 2020-01-01 and tested from that date for every name simultaneously. The frontier survives, with the top decile at +2.47% (per-date DM 6.22 over 1,258 dates; per-name split +2.7%, DM 6.5) and the overall edge +0.71% (DM 1.87). The bottom decile is elevated at +1.3% because a 2020–22 test era leaves few calm cells, but the top decile roughly doubles each other cell. Inside the 2000–2013 holdout, re-estimating strictly before 2007-01-01 and testing on 2007–2013 removes the 2008 crisis from every training window; the pooled learner still wins overall (+0.29%, per-date DM 3.08 over 1,762 dates) and the decile profile keeps its shape (top three deciles +0.45–+1.04% against +0.06–+0.09%). The annual walk-forward refits the per-name GARCH on expanding data at each January-1 cutoff from 2020 to 2024, rebuilds the residuals and score, and retrains the learner; the top decile carries +2.09% at DM 4.43 while the overall edge is no longer detectable (−0.18%, DM −0.52).

Family-wise error over the signal-decile grid. The frontier grid spans three signals and ten deciles. Romano–Wolf step-down family-wise control over the full 3×10 grid takes studentized per-date mean edges per cell, a stationary bootstrap over dates that draws the same dates for every cell (mean block 10 days, B = 2,000, block 20 as sensitivity), and one-sided step-down adjusted p-values. Nine of the thirty cells survive under both block lengths, all frontier cells: the top deciles of the three signals (mk$_{63}$ $t = 9.4$, skew$_{63}$ $t = 9.2$, jump$_5$ $t = 5.1$, adjusted $p < 0.001$), the ninth deciles of mk$_{63}$ (adjusted $p = 0.022$) and skew$_{63}$, skew$_{63}$'s seventh, and jump$_5$'s U-shaped far cells. Mid-grid cells with an unadjusted t near two lose significance under the adjustment.

The point-in-time universe. The holdout's observation filter conditions on names present for most of the era, tilting the universe toward survivors. A point-in-time re-selection fixes the universe at the top 200 names by market capitalization as of Q1-2000, applies no minimum-history filter, and rides each delisting return to the end. On the 164 of 200 names that clear the symmetric fitting floors (36 dropped, 79 delisting returns ridden to the end), the overall edge is +0.58% at per-date DM 8.96 over 2,692 dates, the frozen design-era top bucket carries +2.94% at DM 6.6 (5.9% occupancy), and the bottom bucket is +0.44% (DM 5.4). The edge is larger than in the survivor-tilted holdout, so the observation filter biased the reported edge downward.

OA.7 Estimator schematic and evaluation panels

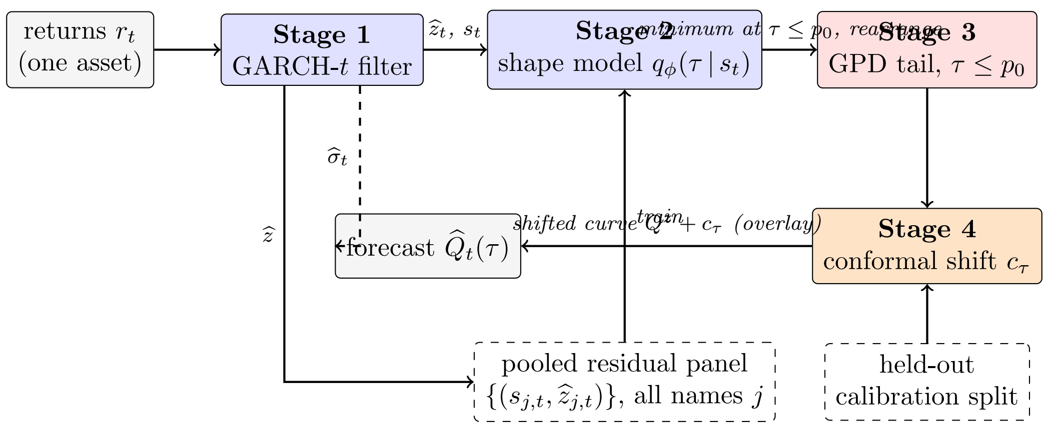

Figure OA.7: The residual-hybrid estimator as a pipeline (Section 2 of the paper). Solid arrows carry data; the dashed top arrow carries the conditional scale $\widehat{\sigma}_t$ that rescales the finished residual quantile curve. Dashed boxes are inputs shared across assets (the pooled panel that trains the amortized shape model) or held out from training (the calibration split behind the conformal overlay).

Table OA.9: Sample flow across evaluation panels (Section 3 of the paper).

Panel	Source	Era	Names	Test obs	Role
Design equity panel	CRSP/WRDS	2014–2024	200	221,600	frontier map, estimator development
Untouched-era holdout	CRSP/WRDS	2000–2013	200	280,608	frozen-spec replication
Exception restatement	CRSP/WRDS	2014–2024	140	≈155,000	per-asset backtesting
Cross-asset sweep	Bloomberg	through 2026	43	[a]	frontier breadth
Country indices	Bloomberg	through 2026	26	[a]	frontier breadth

[a] Frozen exhibits. Per-instrument windows vary and are documented instrument-by-instrument in the repository.

OA.8 The two-regime example

A synthetic boundary check: when the frontier does *not* appear

Before the real data, a simulation with known truth (run once, seed fixed) checks the formal results and, more usefully, delineates when the frontier edge should *fail* to appear. The data-generating process is deliberately favorable to the score: unit-variance Student-t residuals whose degrees of freedom switch between a calm regime ($\nu = 12$) and a fat regime ($\nu = 3$) under a persistent Markov chain; forecasters see the trailing 63-day kurtosis score and compete at $\tau \in \{0.01, 0.025\}$ (80,000 days; scale known, isolating shape). First, the Lemma 1 of the paper identity verifies numerically: the regret integral and the Monte-Carlo regret agree to five decimals (3.5×10^{-5} at $\tau = 0.01$). Second, the conformal shift moves realized coverage toward nominal on the held-out split, as Proposition 1 of the paper promises. Third, and most instructive, the score-binned nonparametric forecaster does *not* beat the fixed shape here: a single $t_{\widehat{\nu}}$ fitted on the mixture ($\widehat{\nu} = 6.8$) is nearly unbeatable at these levels, with total regret under 0.5% and no decile gradient. Two mechanisms explain it, and both teach. (i) By Lemma 1 of the paper regret is *quadratic* in the quantile wedge, and at $\tau \geq 0.01$ the wedge between the fitted compromise and either regime is small; static kurtosis *blending* is a second-order sin. (ii) The sample kurtosis of a t_ν law is a noisy statistic when tails are fat: its usual moment-based asymptotic variance requires a finite eighth moment ($\nu > 8$), and the population kurtosis itself does not exist for $\nu \leq 4$, so sixty-three observations cannot reliably rank regimes that differ only in a static ν. The jump and asymmetry signals, whose top deciles independently survive familywise control, corroborate the ordering, and a tail-index (Hill-type) variant of the score is an obvious robustness extension. For the empirical sections this means the real-data top-decile edge of Section 4 of the paper cannot be an artifact of stationary fat tails being averaged over (that configuration is the one the simulation shows a fixed shape handles) and is consistent with *local* shape breaks a single fitted ν cannot straddle: asymmetry, jump clustering, and residual misfit that co-move with the score. This is the same conclusion the Korea 2026 falsification case reaches from the opposite direction (price crashes are not residual misspecification), and it is why the score combines kurtosis with asymmetry and a jump indicator rather than kurtosis alone. (Simulation script `toy_example.py` ships with the code release.)

OA.9 Trading implications and scenario generation

Trading implications. The universes where the nonparametric tail wins (crisis FX, volatility indices, credit spreads, frontier equities) are the desks where tail-shape-aware sizing and hedging matter most. The *direction* of the trade matters as much as its location: a correct physical-measure tail forecast does not by itself make bought crash insurance profitable, because the crash risk premium can survive conditioning on a correct forecast. The application the evidence supports is the opposite side, selling tail protection, where the model supplies tail control rather than alpha on the mean. The gating experiment provides no evidence that the score times individual crash days, so its role there is continuous tail-risk measurement, not a market-timing signal.

Scenario generation. Because the shape stage carries an exact inverse-CDF sampler, calibrating intractable scenario generators (rough volatility, Hawkes, jump-diffusions) for CCAR/ICAAP-style stress design is a natural extension; it is not developed here.

OA.10 Proofs

The four formal results of Section 2.4 of the paper are stated with proof sketches in the body; the complete arguments, with all regularity hypotheses made explicit, are collected here. Throughout, $\rho_\tau(u) = u(\tau - \mathbb{I}\{u < 0\})$ and $q_F = F^{-1}(\tau) = \inf\{x : F(x) \geq \tau\}$ is the lower generalized inverse.

Proof of Lemma 1 of the paper (pinball regret identity)

Assume $\mathbb{E}|Z| < \infty$, so

$$S(q) = (1-\tau)\int_{(-\infty,q]}(q-z)\,dF(z) \;+\; \tau\int_{(q,\infty)}(z-q)\,dF(z)$$

is finite for every q, and S is convex (an expectation of the convex maps $q \mapsto \rho_\tau(z-q)$). For fixed z, $q \mapsto \rho_\tau(z-q)$ is Lipschitz with constant $\max(\tau, 1-\tau)$, so the difference quotients $[\rho_\tau(Z-q') - \rho_\tau(Z-q)]/(q'-q)$ are bounded by the integrable constant $\max(\tau, 1-\tau)$; dominated convergence justifies differentiating under the expectation wherever the one-sided limits exist. Off the atoms of F,

$$\frac{d}{dq}\,\rho_\tau(z-q) = \mathbb{I}\{z<q\} - \tau, \qquad \Longrightarrow \qquad S'(q) = \mathbb{E}\,\mathbb{I}\{Z<q\} - \tau = F(q^-) - \tau,$$

which equals $F(q)-\tau$ for all but countably many q. Since S is convex, hence absolutely continuous, the fundamental theorem of calculus gives

$$S(q) - S(q_F) = \int_{q_F}^{q}\big(F(u)-\tau\big)\,du,$$

the countable atom set being Lebesgue-null; *no continuity hypothesis on F is needed for this identity.* For nonnegativity, note $q_F = \inf\{x : F(x) \geq \tau\}$ implies $F(u) \geq \tau$ for $u \geq q_F$ and $F(u) < \tau$ for $u < q_F$. If $q > q_F$ the integrand is ≥ 0 on $[q_F, q]$, so the integral is ≥ 0; if $q < q_F$,

$$S(q) - S(q_F) = \int_{q_F}^{q}\big(F(u)-\tau\big)\,du = \int_{q}^{q_F}\big(\tau - F(u)\big)\,du \;\geq\; 0,$$

the integrand again ≥ 0 on $[q, q_F]$. Equality holds iff $F(u) = \tau$ for Lebesgue-a.e. u between q_F and q. Finally, if F is absolutely continuous with density $0 < m \leq f \leq M$ on the interval joining q_F and q, then F is continuous there, so $F(q_F) = \tau$ and $F(u) - \tau = \int_{q_F}^{u} f$; hence $m|u-q_F| \leq |F(u)-\tau| \leq M|u-q_F|$, and integrating over the interval yields $\frac{m}{2}(q-q_F)^2 \leq S(q) - S(q_F) \leq \frac{M}{2}(q-q_F)^2$. $\square$

Proof of Proposition 1 of the paper (split-conformal validity)

Let $k = \lceil (n+1)\tau \rceil$ and assume $1 \leq k \leq n$ so that the k-th order statistic $e_{(k)} = c_\tau$ of the n calibration errors $\{e_1, \ldots, e_n\}$ exists; when $k = n+1$ (i.e. τ within $1/(n+1)$ of 1) set $c_\tau = +\infty$, giving coverage 1. Suppose the $n+1$ errors $e_1, \ldots, e_{n+1}$ (calibration plus the test error $e_{n+1} = z_{n+1} - \widehat{q}(\tau \mid s_{n+1})$)

are exchangeable and almost surely distinct. Then the rank R of e_{n+1} among the pooled $n+1$ values is uniformly distributed on $\{1, \ldots, n+1\}$. By construction $\widetilde{q}(\tau \mid s_{n+1}) = \widehat{q}(\tau \mid s_{n+1}) + c_\tau$, so

$$\{z_{n+1} \leq \widetilde{q}(\tau \mid s_{n+1})\} = \{e_{n+1} \leq c_\tau\} = \{e_{n+1} \leq e_{(k)}\} = \{R \leq k\},$$

the middle equality using a.s. distinctness (with ties the middle equality fails; e.g. if all e_i are equal the event has probability 1, which can exceed the stated upper bound). Therefore

$$\Pr\left(z_{n+1} \leq \widetilde{q}(\tau \mid s_{n+1})\right) = \Pr(R \leq k) = \frac{k}{n+1} = \frac{\lceil (n+1)\tau \rceil}{n+1}.$$

Since $(n+1)\tau \leq \lceil (n+1)\tau \rceil < (n+1)\tau + 1$, dividing by $n+1$ gives $\tau \leq k/(n+1) < \tau + \frac{1}{n+1}$. The probability is over the joint law of calibration and test points; the guarantee is marginal, not conditional on s_{n+1}, because a single scalar shift c_τ pooled across states cannot deliver state-conditional coverage. □

Proof of Proposition 2 of the paper (tail wedge)

Fix $\nu > 2$, so the t_ν law has finite variance and its unit-variance version $Z = T_\nu \sqrt{(\nu-2)/\nu}$ is well defined. The t_ν density is $\sim C_\nu |x|^{-(\nu+1)}$ as $|x| \to \infty$, so the lower tail is regularly varying with index $-\nu$: $\Pr(T_\nu < -x) = \tilde{c}_\nu x^{-\nu}(1 + o(1))$. Unit-variance scaling multiplies the constant, $\Pr(Z < -x) = c_\nu x^{-\nu}(1 + o(1))$ with $c_\nu = \tilde{c}_\nu((\nu-2)/\nu)^{\nu/2}$, and leaves the index ν unchanged. Writing $U(\tau) = |Q_\nu(\tau)|$ and inverting the regularly varying tail (Embrechts et al., 1997) gives $\tau = c_\nu U(\tau)^{-\nu}(1+o(1))$, hence $U(\tau) = (c_\nu/\tau)^{1/\nu}(1+o(1))$ as $\tau \downarrow 0$, i.e. $Q_\nu(\tau) = -(c_\nu/\tau)^{1/\nu}(1+o(1))$. For $\nu_1 < \nu_2$,

$$\frac{|Q_{\nu_1}(\tau)|}{|Q_{\nu_2}(\tau)|} = \frac{(c_{\nu_1}/\tau)^{1/\nu_1}}{(c_{\nu_2}/\tau)^{1/\nu_2}} (1 + o(1)) = \text{const} \cdot \tau^{-(1/\nu_1 - 1/\nu_2)}(1 + o(1)) \longrightarrow \infty$$

because $1/\nu_1 > 1/\nu_2$. Thus for all sufficiently small τ both quantiles are negative with $|Q_{\nu_1}(\tau)| > |Q_{\nu_2}(\tau)|$, i.e. $Q_{\nu_1}(\tau) < Q_{\nu_2}(\tau)$, and the wedge $|Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau)| \to \infty$. The ordering therefore holds for every $\tau < \delta$ for some $\delta > 0$; putting $\tau^* = \min(\delta, \frac{1}{2})$ gives $\tau^* \in (0, \frac{1}{2})$ as asserted. (This establishes *existence* of τ^*; the actual crossover τ_c where the ordering flips is not claimed here and is located numerically in the crossover remark of the paper.) Under local truth t_{ν_1}, applying the corresponding equation of the paper with $q = Q_{\nu_2}(\tau)$ and $q_F = Q_{\nu_1}(\tau)$ bounds the regret below by $\frac{m}{2}(Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau))^2$, with m the minimum of the t_{ν_1} density on the wedge interval. □

Remark on the score ordering

We state the score-ordering as an empirical regularity rather than a theorem, and record here why no clean monotonicity theorem holds at the deployable levels. The tail-wedge result (Proposition 2 of the paper) orders a given pair of Student-t tail indices only below a pair-specific crossover level $\tau^*(\nu_1, \nu_2)$; it does not deliver monotonicity of the wedge in the tail index at a fixed operational τ. More fundamentally, the innovations are unit-variance by construction, and for the unit-variance t_ν the level-τ quantile is $Q_\nu(\tau) = \sqrt{(\nu-2)/\nu}\; t_\nu^{-1}(\tau)$, which is *not* monotone in ν: as $\nu \downarrow 2$ the normalization $\sqrt{(\nu-2)/\nu} \to 0$ pulls the fixed-τ quantile back toward zero, so an arbitrarily heavy-tailed unit-variance t eventually has a fixed-τ quantile arbitrarily close to the center rather than far into the tail. The non-monotonicity is already active at the regulatory levels: at $\tau = 0.025$, $Q_5(0.025) \approx -1.99$ is *less* extreme than $Q_8(0.025) \approx -2.00$, so a heavier-tailed standardized t carries the less extreme 2.5% quantile. A clean ordering theorem would therefore have to restrict the tail index to an explicit interval, isolate a range of τ on which $\partial Q_\nu(\tau)/\partial\nu$ has one sign, and control the density factor in the regret expansion; we do not pursue that narrow statement. The exact content we retain is the regret identity (Lemma 1 of the paper) and the tail-wedge ordering (Proposition 2 of the paper), and the frontier itself is documented empirically in the paper. $\square$

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Proof of Lemma 2 of the paper (generative sampler)

Let $\widetilde{q}(\cdot \mid s)$ be nondecreasing and left-continuous on $(0,1)$ and $\tau \sim U(0,1)$, and set $Z^* = \widetilde{q}(\tau \mid s)$ and $G(x) = \sup\{t \in (0,1) : \widetilde{q}(t \mid s) \le x\}$ (with $\sup \emptyset = 0$). Monotonicity makes $\{t : \widetilde{q}(t \mid s) \le x\}$ an interval with left endpoint 0, and left-continuity ensures the supremum is attained when the set is nonempty, so the set equals $(0, G(x)]$; hence $\{Z^* \le x\} = \{\tau \le G(x)\}$ and $\Pr(Z^* \le x) = G(x)$, i.e. Z^* has CDF G. Because G so defined is the upper generalized inverse of $\widetilde{q}$, and a left-continuous nondecreasing $\widetilde{q}$ is in turn the lower generalized inverse of G, the quantile function of Z^* is exactly $\widetilde{q}(\cdot \mid s)$. When $\widetilde{q}$ is an estimated curve, the argument applies verbatim to the estimate: the sampler is exact for the *fitted* conditional law, which is the claim. $\square$